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Teachers' Preparedness in the Use of Information Communication Technology and School's Licensure Passing Rate

ZENAIDA F. LUCAS
https://orcid.org/0000-0003-3447-9761
zenaidalucas786@gmail.com
Cagayan State University
Andrews Campus
Cagayan, Philippines

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ABSTRACT

Modern-day teaching entails the development of 21st-century skills, which include information and digital literacy, as well as technological skills. All educators face the challenge of keeping abreast of the ever-changing information communication technology and integrating available ICT facilities and tools in teaching. This study was conducted to find the correlation between the teachers' preparedness, perception, and types of ICT facilities and tools used for instruction and the school's licensure passing rate. Descriptive, quantitative, and correlational research designs were adopted with the modified questionnaire of Davis et al. (1989) on teachers' preparedness and perceptions in using ICT, the Technology Acceptance Model Questionnaire (TAMQ) as the main instrument in gathering the needed data. The reliability coefficient of the instrument was established using Cronbach Alpha and yielded indices of 0.84 and 0.99. The questionnaire was administered through Google Form to the 50 teachers in the College of Education at the University of Morong and Antipolo Campus. The study found that female-dominated teachers in the College of Education in the URS are in their early adulthood stage and have been using ICT in teaching for a decade. Most of the teachers have attended one ICT training per year, and teachers are prepared for the use of ICT. An association exists between the teachers' assessment of available ICT facilities and tools and their perception of the use of ICT. The URS performed well in the Licensure Examination for Teachers. Other findings showed that the younger the teachers, the better their perception of the use of ICT; the younger the teachers, the higher their performance in Licensure Examination for Teachers. It can also be concluded that the more prepared the teachers are, the better their school's performance in Licensure Examination for Teachers. Likewise, the more prepared the teachers are, the better their perception of the use of ICT in teaching and vice versa. It can also be concluded that the better the teachers' perception of the use of ICT, the higher the performance in the Licensure Examination for Teachers. The perception of the use of ICT and the performance in the LET is differentiated by age. With this, teachers and administrators must continually support the ICT professional development to keep abreast with the current updates in technology. Preparedness, availability, and perception of the use of ICT must be considered in the school's overall plan, which eventually translates to better school performance.

KEYWORDS

ICT Facilities and Tools, Information Communication Technology, Licensure Examination for Teachers, Teachers' Perception, Teachers' Preparedness, descriptive correlational, Philippines

INTRODUCTION

Teachers' competencies are expanding to address the needs of 21st-century learners. This makes the teaching job evolving and challenging. Teachers must be more competent in using technology than their 'digital native' students, develop 21st-century skills, and develop new teaching and learning methods (UNESCO, 2010).

Educators need Technological Pedagogical Content Knowledge (TPACK), Koehler and Mishra (2006) to efficiently use technology in the teaching and learning process. Rodgers (2018) explained that Content Knowledge (CK) is the teachers' knowledge of the subject matter. Pedagogical Knowledge (PK) is the teachers' knowledge of the students and strategies to meet their needs. Technological Knowledge (TK) refers to technological knowledge with proper tools for the lesson and the students.

Education graduates who aspire to teach in the Philippine public school system must pass the Licensure Examination for Teachers (LET) (Apare et al., 2018). The Philippine Regulatory Commission checks the licensure examinations of the different regulated professions, which include the teaching profession. Philippine Congress, through the Senate and the House of the Representatives, enacted several laws such as Republic Act no. 9293, amending sections of R.A. No. 7836, 'Philippine Teachers Professionalization Act of 1994, R.A. 8981, Modernization Act of 2000, Section 7(m).

To monitor the teacher competencies for the teaching profession, the Philippine Qualifications Framework (PQF) was created under Republic Act No. 10968. The PQF Act states that it is the state's policy to institutionalize the PQF to encourage lifelong learning of individuals aligned with the industry standards. The quality services must be aligned with the Philippine Qualifications Framework (PQF)for national relevance, global compatibility, and competitiveness.

Teaching in the new normal poses a great challenge of incorporating skills and competencies in using Information Communication Technology in teaching and learning.

FRAMEWORK

This study focused on the Theory of Reasoned Action (TRA) (Fishbein, 1967). TRA was developed to understand the relationship between attitudes, intentions, and behavior. Davis (1986) adapted TRA to address user acceptance of information systems in his Theory of Acceptance Model (TAM). TAM predicts individual adoption and use of modern technologies in a work context based on perceived usefulness (PU) and Perceived Ease of Use (PEOU). Perceived Usefulness is the degree to which a person believes that work performance can be increased using the system. Perceived Ease of Use (PEOU) is the degree to which the system will be effortless.

OBJECTIVE OF THE STUDY

This study aimed to find the correlation between teachers' preparedness in the use of ICT to the School's Licensure Examination passing rate.

METHODOLOGY

Research Design

This study used descriptive and correlative design. It is descriptive since it describes the study's teacher respondents according to their age, sex, years of using ICT, and the number of ICT trainings attended. These profile variables were correlated to the teachers' preparedness, perception of the use of ICT, ICT facilities and tools, and their school's Licensure Examination Passing Rate.

Moreover, the study found whether there is any significant relationship between the teachers' use of ICT and their perception, teachers' use of ICT and Licensure Passing Rate, teachers' profile, and their level of preparedness in the use of ICT.

Research Locale and Respondents

This study was conducted on two campuses in the College of Education in the University of Rizal System in the province of Rizal. These campuses are URS Morong and URS Antipolo. The respondents of the study were the 50 teachers from URS Morong and URS Antipolo.

Research Instrument

The researcher used a structured questionnaire for the teachers' profiles and ICT facilities and tools. To determine the teachers' preparedness, a structured questionnaire using a 5-point Likert Scale was used; 1 Not Prepared, 2 Slightly Prepared, 3 Moderately Prepared, 4 Prepared, and 5 Highly Prepared.

The researcher adopted and modified Davis et al. (1989) Technology Acceptance Model Questionnaire (TAMQ) to find out the teachers' Perceived Use (PU) and Perceived Ease of Use (PEOU) using a 5-point Likert Scale; 1 Not Useful, 2 Slightly Useful, 3 Somewhat Useful, 4 Useful, and 5 Very Useful.

Data Gathering Procedure

The researcher sent a letter to the Data Privacy Officer of the University of Rizal System. The approved letter was endorsed to the Office of the University President for approval. The President's approval letter was forwarded to the Dean of the College of Education. Then, the COE secretary forwarded the Google Form questionnaire to the teachers' email addresses. The data were summarized and submitted to the statistician for analysis and interpretation.

Statistical Treatment

This study used proper statistical tools to analyze and interpret the data gathered. Frequency counts, means, and percentages were used to find the respondents' profiles. To find the level of teachers' preparedness, frequency, percentage, and weighted mean were used. To find the level of teachers' perception, frequency, percentage, and weighted mean were also used. To determine whether there is a significant relationship between the factors: the teachers' use of ICT facilities, tools, and perception; teachers' profile and level of preparedness, school licensure, and their perception of the use of ICT, the Pearson R, and Chi-square values were used.

To find the difference in the teachers' level of preparedness in the use of ICT, the level of teachers' perception in the use of ICT, and the school's licensure passing rate when grouped according to their profile, the researcher used ANOVA and T-test.

RESULTS AND DISCUSSION

The following were the salient findings of the study: the majority of the respondents were females 21-35 years old. Teachers have attended one ICT training per year. Teachers were prepared to use ICT. There is a significant difference between teachers' use of available ICT facilities and tools and their perception of the use of ICT.

Based on March 2022 Licensure Examination for Teachers, the combined LET performance of the two campuses showed that their Bachelor of Secondary Education (BSE) program registered a passing rate far higher than the national passing rate. In the Bachelor of Elementary Education (BEED) program, the school's LET is higher than the national passing rate.

There is a highly significant relationship between the teachers' perception of the use of ICT and the school's Licensure Passing Rate. If teachers have a positive outlook on using ICT in their classes, this results in a high Passing Rate in their school's Licensure Examination for Teachers.

When grouped according to the teachers' profile variables, Age is the only factor with a significant difference in the level of teachers' perception on the use of ICT.

CONCLUSIONS

The female-dominated teachers in the College of Education are in the early adulthood stage in life and have been using ICT for almost a decade. The majority of the teachers attended one ICT training per year.

The ICT facilities and tools are available for instruction. The respondents are prepared to use these facilities. Teachers perceived ICT to be very useful in their work performance.

Based on March 2022 Licensure Examination for Teachers, the University of Rizal System performed well above the national passing rate.

It also concluded that the younger the teachers, the better their perception of the use of ICT; the younger the teachers, the higher their performance in Licensure Examination for Teachers.

The more prepared the teachers are, the better the performance in LET. Likewise, the more prepared the teachers are, the better their perception of using ICT and vice versa.

It also concluded that the better the teachers' perception of using ICT, the higher the performance in LET.

Lastly, the perception of the use of ICT and the performance in the LET is differentiated by age.

RECOMMENDATIONS

Based on the findings and conclusions of the study, the following recommendations are given:

The school administration must conduct continuous ICT trainings and technical support to integrate ICT into teaching and learning for teachers.

The administration must provide trainings on the use of ICT facilities and tools to prepare teachers and develop a higher perception of the use of ICT for teaching and learning.

Future researchers on preparedness and perception of the use of ICT may include the pre-service teachers to rate the usefulness of ICT in instruction as well as the preparedness of the teachers as a form of triangulation of the data gathered.

LITERATURE CITED

Adelman, C., Ewell, P., Gaston, P. & Schneider, C. G. (2014) Degree Qualification ProLumina Foundation.

Alimyar, Z., Lakshmi, G. and Sanchez-Ramos, M. (2021). A study on language teachers'

preparedness to use technology during COVID-19 Cogent Arts & Humanities (2021), 8: 1999064.

Alvarez, A. Jr. (2021). Rethinking digital divide in the time of crisis

Apare, B., Arcilla Jr, F., & Vasquez, O. (2018). Academic achievement in the licensure examination for teachers of education graduates. IAMURE Multidisciplinary Research, 17(1), 14-25.

Bader, S. (2021). The Impact of Instructors' Perceptions of E-Learning on the Quality of Online Teaching: A Case Study of the French Language Instructors at the University of Bahrain During COVID-19 Pandemic.

Barium, Iddrisu (2019). Influence of Teachers' Gender and Age on the Integration of Computer Assisted Instruction in Teaching and Learning of Social Studies Among Basic schools in Tamale Metropolis.

Banal, I. (2015). Teachers as digital immigrants. Sunstar Pampanga.

Bhandari, P. (2021) What is Quantitative Research? Definition, Uses and Methods

Beyle, C. (2015). Learning in virtual environments: an integrative approach for understanding the adoption, engagement and learning achievement in digital context.

Buabeng-Andoh, C. (2012). Factors influencing teachers' adoption and integration of information

and communication technology into teaching: A review of the literature. International Journal of Education and Development using ICT, 8(1). Contouring Quality in Education. AT: Mumbai,

Creswell, J. W. (2013). Research design: Qualitative, quantitative, and mixed method approaches. Sage publications.

De Los Angeles, M.A.G. (2019. Predictors of Performance in Licensure Examination for Teachers, 2020, Universal Journal of Educational Research, Horizon Research Publishing (HRPUB)

Fadip, N., Yusahu, B., Ya'u, Gital A. (2018). Lecturer's Level of awareness of ICT Facilities for Teaching Purposes

Dimaggio P, Hargittai E (2001) From the "digital divide" to "digital inequality": studying Internet use as penetration increases. Center for Arts and Cultural Policy Studies. Woodrow Wilson School, Princeton University.

D. Amutha 2020, The Role and Impact of ICT in Improving the Quality of Education. St. Mary's College, Thoothukudi, Tamilnadu, India 628001

Erasmus, E., Rothmann, S., & Van Eeden, C. (2015). A structural model of technology acceptance. SA Journal of Industrial Psychology/SA Tydskrif vir Bedryfsielkunde, 41(1), Art. #1222, 12 pages.

Fitzallen, N., and Brown, N. (2005). What Profiling Tell Us about ICT and Professional Practice.

Fleming, L., Motamedi, V., & May, L. (2007). Predicting Preservice Teacher Competence in Computer Technology: Modeling and application in Training Environments.

Flynn, S. (2021). Education, Digital Natives, and Inequality.

Frankowicz, M. (2016). New ICT and changing roles of teachers. The Journal of Quality Education, 6(8), 4.

Fulton, J. (2019) 7 Examples of Innovative Educational Technology.

Geeraerts, K, Vanhoof, J., Van Den Boosche, P. (2019). Flemish Teachers' Age-related Stereotypes: Investigating Generational Differences.

Gurung, Lina. (2016). Need of Digital Competency among Female Teachers.

Fitzallen, N. (2004). Profiling Teachers' Integration of ICT into professional practice. Paper presented at AARE Annual Conference, Melbourne.

Hero, J. L. (2020). Teachers' Preparedness and Acceptance of Information and Communication Technology (ICT) Integration and Its Effect on their ICT Integration Practices. Puissant. A Multidisciplinary Journals, 1, 59-76

Jatileni, Malakia & Jatileni, Cloneria N. (2018). Teachers' perception on the use of ICT in Teaching and Learning: A Case of Namibian Primary Education. Master Thesis Philosophical Faculty School of Applied Educational Science and Teacher Education

Jegede, Phillip Olu. (2009). Age and ICT-Related Behaviours of Higher Education Teachers in Nigeria.

Kemp, S. (2021). Digital 2021: The Philippines Kundu, A., Bej, T and Dey, K.N. (2022) Subject-Self Affecting on Teachers.

Keržič, D., Danko, M., Zorko, V., & Dečman, M. (2021). The effect of age on higher education teachers' ICT use. Knowledge Management & E- Learning, 13(2), 182–193.

Koehler, M.J., & Mishra, P. (2009). What Is technological pedagogical content knowledge? Contemporary Issues in Technology and Teacher Education (CITE), 9(60-70).

Lewis, J. (2018). Comparison of Four TAM Item Formats: Effect of Response Option Labels and Order. Journal of Usability Studies, Vol.14, Issue 4, August 2019, pp. 224-236

Llyod, Margaret, Downes, T. and Romeo, G. (2016). Positioning ICT in Teachers' Career Path: ICT Competency as an Integral Part of Teacher Standards: UNESCO-ICT-CFT-2011

Macale, A., Quimbo, J. D.J. (2019).Digital immigrants teaching digital natives A call for a paradigm shift.

Mahdi, H. S., and Al-Dera, A.S. (2013). The Impact of Teachers' Age, Gender and

Experience on the Use of Information and Communication Technology in EFL Teaching.

Martin, E.L. (2011). Digital natives and digital immigrants: Teaching with technology.

Mansoor, M., Abbas, G., Baloch, G.H. (2017). Use of ICT Facilities in Higher Education

May, M. (2020). Comparison Study Between Math Test scores for Brick and Mortar and online.

Mendoza, R. (2022). Laguna, NTC graduates tops licensure exam.

Mohamud, Mohamed & Ali, Abdifatah. (2018). Evaluating ICT Preparedness of

Higher Education in Somalia.

Morrison, C. (2014). From 'Sage on the Stage' to 'Guide on the Side': A Good Start

Muijs, Daniel (2010). Doing Quantitative Research in Education with SPSS. 2nd edition. London: SAGE Publications.

Nocar, D., Dofkova, R., and Pastor, K. (2019). Primary school teachers' preparedness to develop pupils' digital literacy in teaching mathematics

Ode, J.O. Eriba, J.O. and Abah, J.C. (2020) College of Education Lecturers Awareness of ICT Facilities for Teaching in the Post COVID-19 Era in Benue State-Nigeria, Public Health Lett., 5 (11), Pp.6

Olatunde, Funsho, 2019.An Assessment of Information Communication Technology (ICT) Role in The Teaching And Learning Of Social Studies In Lagos State Schools

Oldfiled, A. (2010). A summary of teacher attitudes to ICT uses in schools

Patankar, P.S., Jadhav, M.S., (2014). ICT skills of teacher for teaching-learning process in higher education. Shivaji University, Kolhapur.

Prensky, M (2001a) Digital Natives, Digital Immigrants: Part 1. On the Horizon, 9(5), 1 - 6.

Prensky, M. (2001b) Digital Natives, Digital Immigrants Part 2: Do they really think differently? On the Horizon, 9(6), 1 – 6.

Rahayu, R.P. and Wirza, Y. (2020). Teachers' perception of online learning during Pandemic-COVID-19.

Rappler (2022). Results: January 2022 Licensure Examination for Teachers

Rodgers, D. (2018) The TPACK framework explained (with classroom examples)

Romeo, G. (2016). "Positioning ICT in teachers' career path: ICT competency as an integral part of teacher's standards (Australia) (Chapter 3) (2016)" Published in 2016 by the United Nations Educational, Scientific and Cultural Organization and UNESCO Bangkok Office

Sapna, Gupta, A. & Shukla, S. (2021) Satisfaction of Male and Female Teachers in Indian Higher Education: A Review

Schiler, J. (2003). Working with ICT: Perceptions of Australian principals, Journal of Educational Administration, vol. 41, no. 3, pp. 171-185.

Shrivastav, P. & Garg, I. (2015). Role of teacher in preparing quality teachers in ICT. Conference:

Ungar, O. & Baruch, A. (2016). Perceptions of Teacher Educators Regarding ICT Implementation in Israeli Colleges of Education. Interdisciplinary Journal of E-Learning and Learning Objects, 12(1), 279-296. Informing Science Institute.

Viberg, O., Balter, O., Mavroudi, A., & Khalil, M., (2020). Validating an Instrument to Measure Teachers' Preparedness to Use Digital Technology in their Teaching, Nordic Journal of Digital Literacy,

Wheeler, Steve. (2001) Information and Communication Technologies and the Changing Role of the Teacher, Journal of Educational Media, 26:1, 7- 17, DOI: 10.1080/1358165010260102

Yukselturk, E., & Bulut, S. (2009). Gender differences in self-regulated online learning environment. Journal of Educational Technology & Society, vo1.2, no.3, pp.12-22

Yusri, L., Muni, C., Goodwin, R. (2016) Teachers and ICT: Towards an effective Training for Teachers